

Chapter 2: Credit Creation by Commercial Banks

1. Introduction and Meaning

- Credit creation is one of the most vital functions performed by commercial banks.
- It refers to the process by which commercial banks expand their total deposits and money supply through the mechanism of accepting public deposits and advancing loans.
- Through this process, banks create credit that is many times more than their initial cash reserves.

2. Basic Assumptions of Credit Creation

To understand how credit creation works theoretically, economists make two key assumptions based on historical banking behavior:

- **The Banking System as a Whole:** The process is analyzed for the entire banking system, treating all commercial banks as a single consolidated unit.
- **All Transactions are Routed Through Banks:** All payments and receipts in the economy are made through bank transfers, meaning that money lent out eventually finds its way back into the banking system as a new deposit.

3. Key Concepts Involved

- **Initial Deposits (Primary Deposits):** The actual cash deposited by the public into commercial banks.
- **Legal Reserve Ratio (LRR):** The minimum fraction of total deposits that banks are legally required to keep as liquid cash, mandated by the central bank.
- **Derivative Deposits (Secondary Deposits):** The deposits created indirectly when a bank grants a loan to a borrower, which is credited to their account.
- **Credit Multiplier (or Money Multiplier):** The multiple by which total deposits increase due to a change in initial reserves. It is inversely related to the LRR:

$$\text{Multiplier}(k) = \frac{1}{\text{LRR}}$$

4. Step-by-Step Numerical Example and Complete Table

Let us assume:

- **Initial Cash Deposit:** \$1,000
- **Legal Reserve Ratio (LRR):** 20% (or 0.2)

Complete Multi-Round Breakdown Table:

The process continues through successive rounds until the cumulative reserves equal the initial deposit, multiplying the total money supply:

Round	Deposits Received	Loans Advanced	Cash Reserves (20%)
Round 1	1,000.00	800.00	200.00
Round 2	800.00	640.00	160.00
Round 3	640.00	512.00	128.00
Round 4	512.00	409.60	102.40
Round 5	409.60	327.68	65.54
Round 6	327.68	262.14	52.43
Round 7	262.14	209.72	41.94
Round 8	209.72	167.77	33.55
Round 9	167.77	134.22	26.84
Remaining Rounds (Converging to Zero)	670.93	536.73	134.20
Total	5,000.00	4,000.00	1,000.00

Calculation using the Multiplier Formula:

$$\text{Total Credit Creation} = \text{Initial Deposit} \times \frac{1}{\text{LRR}}$$

$$\text{Total Credit Creation} = 1,000 \times \frac{1}{0.2} = 1,000 \times 5 = 5,000$$

Thus, an initial cash deposit of 1,000 expands into a total money supply of 5,000 in the economy.

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6. Limitations on Credit Creation

The actual volume of credit created by commercial banks may be lower than the theoretical maximum due to several constraints:

- **High Legal Reserve Ratios:** If the central bank increases the LRR, the credit multiplier shrinks, reducing the capacity of banks to create credit.
- **Cash Drain (Currency Leakage):** If the public prefers to hold a portion of their loans or transactions in physical cash rather than routing everything through bank accounts, the multiplier effect slows down.
- **Availability of Credit-Worthy Borrowers:** Even if banks have surplus reserves to lend, credit creation will fall short if there is a lack of qualified borrowers willing to take loans for investment or consumption.
- **Bank's Excess Reserves:** If commercial banks choose to hold reserves higher than the legal minimum for safety, less money is available for lending.

7. Significance in the Economy

- **Boosts Liquidity:** It ensures adequate money supply and liquidity required to match growing levels of trade and commercial transactions.
- **Capital Formation:** The lending activities channel idle savings into productive investments, encouraging industrial and agricultural growth.
- **Economic Development:** By facilitating credit for business expansions, housing, and infrastructure, commercial banks directly contribute to national economic progress.

2. High Powered Money

1. Meaning of High Powered Money

High Powered Money (HPM) refers to the total amount of money created by the central bank of a country. It is also known as:

- Reserve Money
- Base Money
- Monetary Base
- Central Bank Money

In India, High Powered Money is created and controlled mainly by the Reserve Bank of India (RBI). It forms the foundation of the entire money supply system because commercial banks use this money as a base for creating credit.

It is called “high powered” because a small increase in this money can create a much larger increase in the total

money supply through the process of bank lending and credit creation.

3. Why is it called High Powered Money?

High Powered Money receives this name because it has the ability to generate multiple times more money through the banking system.

Suppose RBI issues ₹100 crore as additional reserve money. Banks receive this additional reserve and keep only a small portion as required reserves. The remaining amount is given as loans. The loan amount again comes back into banks as deposits. Banks again lend a part of those deposits. This repeated process creates multiple deposits and increases total money supply.

Therefore:

Increase in High Powered Money



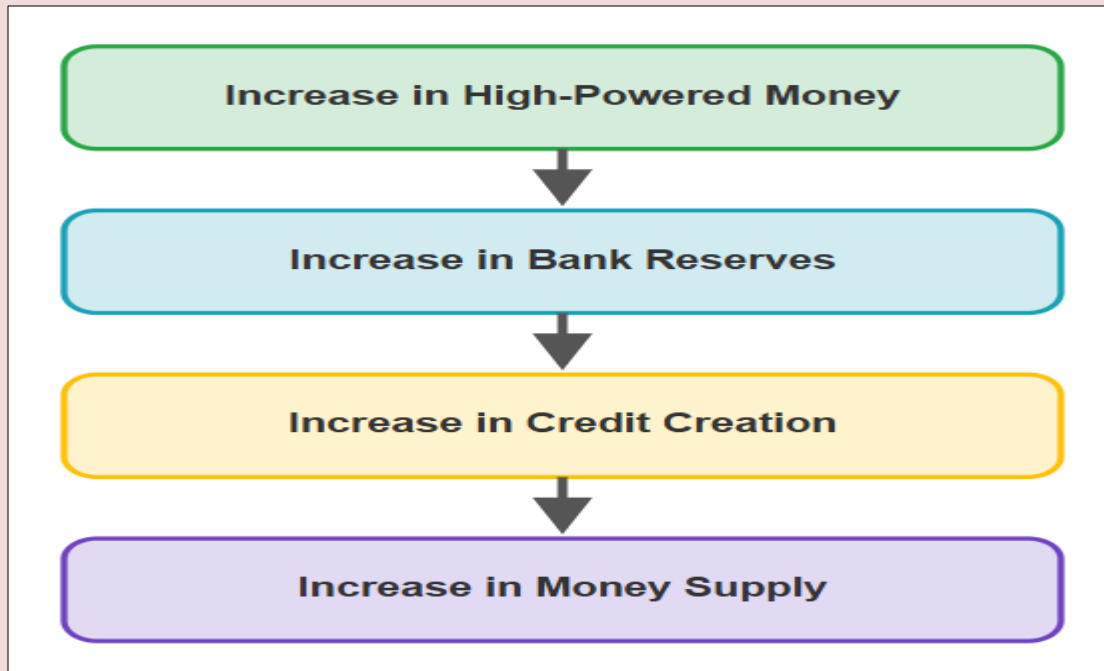
Increase in Bank Reserves



Increase in Credit Creation



Increase in Money Supply



4. Components of High Powered Money

High Powered Money consists of three main components:

$$\mathbf{H = C + R + OD}$$

Where:

H = High Powered Money

C = Currency held by the public

R = Cash reserves of commercial banks with RBI

OD = Other deposits with RBI

Component 1: Currency with the Public (C)

Currency with the public includes:

- Notes and coins held by individuals
- Currency used for daily transactions
- Cash available outside the banking system

Example:

Money kept in your wallet or cash kept by businesses for transactions forms part of currency with the public.

Component 2: Bank Reserves with RBI (R)

Commercial banks are required to maintain certain reserves with RBI.

These reserves include:

1. Cash Reserve Ratio (CRR)

Banks must keep a certain percentage of their deposits with RBI as cash reserves.

Example:

If a bank has deposits of ₹1000 crore and CRR is 4%, then:

Reserve with RBI = ₹40 crore

2. Excess Reserves

Banks may keep additional reserves above the required limit.

These reserves help banks meet sudden withdrawal demands.

Component 3: Other Deposits with RBI (OD)

Other deposits include deposits maintained with RBI by:

- Financial institutions
- Foreign central banks
- International organisations

These deposits form a small part of total high powered money.

6. Relationship Between High Powered Money and Money Supply

High Powered Money is the **base of money creation**.

The relationship is:

Where:

M = Total Money Supply

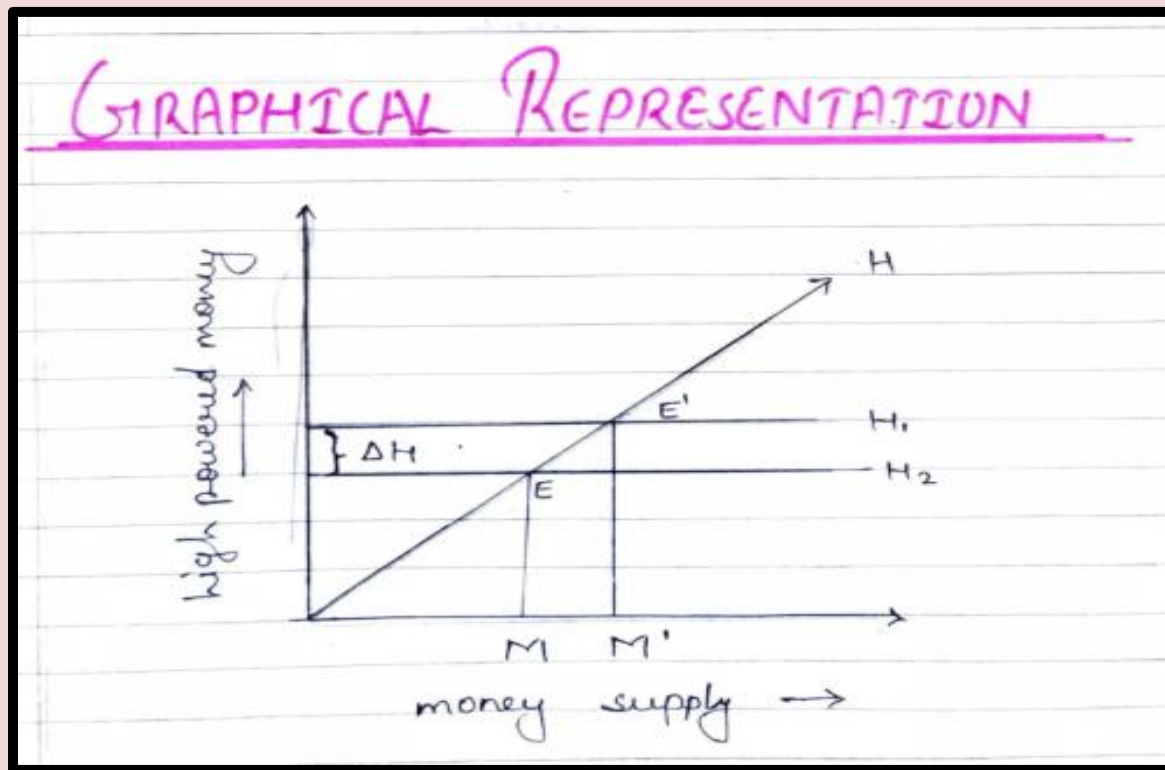
m = Money Multiplier

H = High Powered Money

This means:

Money Supply depends on:

1. Amount of High Powered Money created by RBI
2. Ability of banks to create credit



3. The Money Multiplier

The money multiplier (m) measures the maximum amount of money the banking system can create with each unit of change in high-powered money (the monetary base, H).

- **Simple Money Multiplier** (assumes no public currency holding and zero excess reserves):

$$m = \frac{1}{rr}$$

(where rr is the required reserve ratio)

- **Comprehensive Money Multiplier** (accounts for public currency drain and commercial bank excess reserves):

$$m = \frac{1 + c}{r + e + c}$$

Where:

- $c = \frac{C}{D}$ (Currency-to-deposit ratio)
- $r = \frac{R}{D}$ (Required reserve ratio)
- $e = \frac{ER}{D}$ (Excess reserve ratio)

Suppose we are using the comprehensive money multiplier framework with the following parameters:

- **Increase in High-Powered Money (ΔH):** ₹100 crore
- **Required Reserve Ratio (r):** 10% (0.10)
- **Currency Ratio (c):** 20% (0.20)
- **Excess Reserve Ratio (e):** 5% (0.05)

Step A: Calculate the Money Multiplier (m)

$$m = \frac{1 + 0.20}{0.10 + 0.05 + 0.20}$$

$$m = \frac{1.20}{0.35} \approx 3.43$$

Step B: Calculate the Total Change in Money Supply (ΔM)

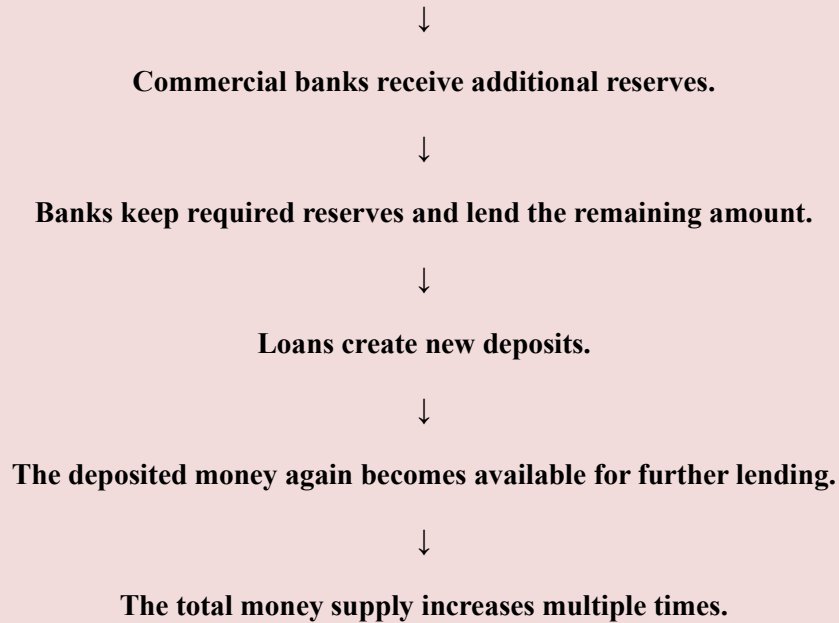
$$\Delta M = m \times \Delta H$$

$$\Delta M = 3.43 \times ₹100 \text{ crore} \approx ₹343 \text{ crore}$$

8. Process of Credit Creation by Commercial Banks

The process works as follows:

RBI creates ₹100 crore new reserve money.



9. Factors Affecting Money Multiplier

The size of money multiplier depends on several factors:

1. Reserve Ratio

Higher reserve requirements reduce money creation.

Lower reserve requirements increase money creation.

Example:

Higher CRR → Less lending → Lower money multiplier

Lower CRR → More lending → Higher money multiplier

2. Currency Holding Behaviour of Public

If people keep more cash with themselves:

- Bank deposits decrease
- Lending capacity decreases

- Money multiplier falls

If people deposit more money in banks:

- Banks receive more funds
- Credit creation increases
- Money multiplier rises

3. Banking Habits

A developed banking system increases the money multiplier because people prefer deposits and electronic payments instead of holding cash.

4. Excess Reserves Maintained by Banks

If banks keep extra reserves instead of lending:

- Credit creation decreases
- Money multiplier decreases